Problem-Solution Fit

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters – AI-Powered Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS]

Disaster Management Authorities & Emergency Coordinators

Meteorological Researchers & Policy Makers

Municipal Councils & Local Safety Agencies

2. PROBLEMS / PAINS

Existing flood prediction systems lack timely and accurate prediction. Physics-based simulations take hours to run, leaving insufficient time for local emergency evacuations.

3. TRIGGERS TO ACT[TR]

Unprecedented heavy monsoon cycles and sudden storm surges.

Impending local flash flood threats.

4. EMOTIONS[EM]

Before

Confused

Time-consuming Difficult

After

Easy

Fast

Informative

6. CUSTOMERLIMITATIONS

Lack of advanced hydrological expertise. Limited local funding for real-time sensor grids. Need a fast, simple prediction interface.

9. PROBLEM ROOT / CAUSE [RC]

Lack of automated, real-time prediction and alert platforms. Complex hydrodynamic modeling calculations. High data processing latency.

10. YOUR SOLUTION [SL]

5. Rising Waters – Machine Learning-Based Flood Prediction and Early Warning System providing real-time, high-accuracy risk classification and alert notifications.

Pros

Physics-based hydrodynamic modeling is highly reliable.

Satellite meteorological reports provide detailed visual analysis.

Cons

Very slow computation and high rendering latency (takes hours).

No automated real-time local classification and interactive dashboard alerting.

7. BEHAVIOR[BE]

Monitoring precipitation patterns, tracking seasonal rain, and analyzing cloud cover density.

Reviewing historical regional flood warning safety thresholds.

8. CHANNELS OF BEHAVIOR

[CH]ONLINE: Interactive Web Applications, SMS Mobile Alerts, Regional Warning Portals

OFFLINE: Local Emergency Warning Sirens, Disaster Operations Radio Broadcasting